Exam 01 Review — Modules 1–6 (Comprehensive Study Session)

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you will be able to:

1. Connect wide-ranging concepts: Atoms → Molecules → Cells → Ecosystems (Modules 1-6).
2. Visually represent biological structures and processes through scientific drawing.
3. Collaborate with peers to explain and clarify complex topics.
4. Identify specific areas for further study before the exam.

Overview

This lab session is dedicated to reviewing Modules 1-6 in preparation for Exam 01. It is a paper-based, collaborative study session designed to help you connect the dots between the "Big Picture" of life, the molecular building blocks, and the cellular machinery.

Materials

- Review Packet (this document)
- Course Textbook/Notes
- Pencils and Colored Pencils

Part 1: The Big Picture (Module 1)

Theme: Life is organized, requires energy, responds to the environment, and evolves.

Draw a diagram showing the hierarchy of life from simplest to most complex: Atom → Molecule → Organelle → Cell → Tissue → Organ → Organism → Population → Ecosystem.

1. Select one arrow from your diagram. Explain how the lower level (e.g., molecule) determines the function of the higher level (e.g., organelle).

Part 2: The Chemistry of Life (Modules 2 & 3)

Theme: Life is based on chemical rules. The shape of a molecule determines its function.

Box A: Draw two water molecules and show the Hydrogen Bond between them.
Box B: Draw a simple representation of an Enzyme (Protein) binding to a Substrate.

Label: Active Site, Substrate, Product

2. How does the polarity of water (Part A) relate to the folding of the protein (Part B)? (Hint: Hydrophobic vs. Hydrophilic)

Part 3: The Cell & Its Systems (Modules 4 & 5)

Theme: Cells are the fundamental units of life. They manage matter, energy, and information.

Draw a large Eukaryotic cell. Include and label:

- **Nucleus** (The Manager/Library)
- **Mitochondria** (Power Plants)
- **Chloroplasts** (Solar Panels - if plant cell)
- **Ribosomes** (Assembly Lines)
- **Cell Membrane** (Security Fence)

3. Trace the path of a protein that is going to be secreted from the cell. Where is it made? Where does it go next? How does it leave?

4. Compare Passive Transport (Diffusion) with Active Transport. Which one needs energy (ATP)? Why?

Part 4: Energy & Metabolism (Module 6)

Theme: Energy flows through systems while matter cycles.

Draw a diagram connecting a Chloroplast and a Mitochondrion. Show the flow of energy and the cycling of matter. Include and label:

- **Chloroplast**
- **Mitochondrion**
- **Light Energy**
- **ATP (Cellular Energy)**
- **Glucose, O_2, CO_2, H_2O**

5. Cellular respiration produces ATP. What is ATP, and why does a cell need it?

6. Describe how the products of photosynthesis serve as the reactants (ingredients) for cellular respiration.

Conclusion

Biology is not just a list of facts; it is a web of connections.

One concept I feel very confident about is:

One concept I need to study more is: